

The Effect of Contamination on Contamination Limit Regulation

US Coal Mining and Drinking Water Utilities

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Self-reported compliance

- Regulatory models requiring entities to self-report compliance:
 - ▶ Healthcare HIPAA violation disclosure; Energy firms self-report anti-competitiveness/negligence to the Federal Energy Regulatory Commission; Public companies' SOX self-report financial statement accuracy; Drinking water utilities' water quality.
- Incentive to shirk reporting. Consider drinking water utilities' federal water quality regulation:
 - ▶ No self-report $\Rightarrow$ no contamination exceedance violation issued (Naylor, 2024; Zou, 2021).
 - ▶ No self-report saves self-reporting costs (Oleckno, 1982).
- **Incentives rise with contamination**
 - ▶ $\uparrow$ contamination $\rightarrow \uparrow$ testing requirements ($\uparrow$ **reporting costs**)
 - ▶ $\uparrow$ contamination $\rightarrow \uparrow$ probability contamination violation ($\uparrow$ **need to conceal**).
- Whether it is self-reporting costs or concealment has different policy implications for improving reporting.

Drinking water quality and coal mining

- Drinking water utilities provide treated water to 85% of US households.
 - ▶ Utilities source from groundwater and/or surface water, serve from twenty-five to millions of customers, publicly or privately owned.
- 9 to 34 million US households obtain water from a utility violating a health standard (Allaire, Wu, and Lall, 2018).
- **Coal mining is a source of water contamination**
 - ▶ **Acid mine drainage:** sulfuric acid formed from sulfide rock dissolves metal ions and sulfates, travels into surface and ground water (Skousen, Ziemkiewicz, and McDonald, 2019).
 - ▶ **Coal wash slurry:** impurities stored in retaining ponds seep through porous rock, or ponds fail.
- Federal policy is expanding mining access and delaying plant retirements (U.S. President (Trump) (2025); Daly (2025)).
- Coal supplied 16% of US electricity in 2025 (International Energy Agency (2025)).

Research questions

Does coal mining raise contamination at drinking water utilities?

What happens to firm self-reporting when contamination rises exogenously, under self-reported compliance?

Is any reduction in self-reporting concealment of exceedances, or avoidance of reporting costs?

What I do

- Setting: drinking water utilities downstream of coal mines, 1985–2005.
 - ▶ Contamination is an **upstream externality** — the firm cannot reduce it by cutting output (Benjamin, 2005).
 - ▶ Separating contamination from utility choices helps identify the effect of contamination.
- Adapt Douglas and Wiggins (2015) instrument for coal production to address two sources of endogeneity:
 - ▶ Coal mines are near coal seams and transportation that independently predict water quality and utility characteristics.
 - ▶ Regulators block upstream mining for vulnerable utilities.
- Contributions:
 - ▶ Causal evidence that contamination reduces self-reporting.
 - ▶ Rationalize utility decisions to self-report in light of regulator's actions with a model.
 - ▶ First evidence that **the cost of self-reporting**, not only concealment, causes environmental under-reporting (Naylor, 2024; Zou, 2021).
 - ▶ First causal estimates of the effect of **coal mining on treated drinking water** at US utilities.

What I find

- ↑ mining → ↑ contaminant concentrations, but never above the MCL. 10M cumulative tons upstream raises mean nitrate by 11.0% of its sample mean (***) and selenium by 8.5% (**); arsenic and barium show no significant change.
- ↑ mining → ↑ **under-reporting**. One more upstream mine raises MR violations by 3–5 pp, against base rates of roughly 4–6%.
- ↑ mining → **no significant change** in MCL violations.
- ↑ mining → ↑ **inspection visits** (+2.1 pp).
- Formal enforcement: −3.4 pp with utility and year fixed effects, −1.1 pp (not significant) with state × year fixed effects, against a base rate of 3%.
- **Self-reporting costs, not concealment:**
 - ▶ When nitrate readings cross 50% of the MCL → monitoring requirements ↑ (reporting cost ↑) → self-reporting ↓.
 - ▶ Sanitary inspections at under-reporting utilities do not discover concealed violations.
 - ▶ Contaminant concentrations stay below MCL.

Contamination shifts monitoring from the firm onto the regulator.

Self-reporting model

Institutional setting

- Utilities federally regulated by the Safe Drinking Water Act (“SDWA”).
- Contaminants cannot exceed a Maximum Contaminant Level (“MCL”).
- Self-report to regulator or get a Monitoring and Reporting violation (“MR”).
- Self-reporting requirements depend on water contamination/threats to water.
- States are SDWA enforcers with discretion over enforcement and visits:
 - ▶ Sanitary survey - comprehensive on-site inspection
 - ▶ Technical assistance - helps fix problem or become compliant.
 - ▶ Sampling - prompted by suspected contamination event or violation.
 - ▶ Enforcement visit - triggered by suspected violation.
- Enforcement, while remedial rather than punitive, is still costly:
 - ▶ Informal (letters, notices, compliance plans)
 - ▶ Formal (consent decrees, fines, prosecution)
 - ▶ MR violations are punished less severely than MCL violations.

Model: the utility's decisions to self-report or under-report

Utility draws compliance cost type $\theta \sim F(\cdot; m)$, worsening in upstream contamination m .

- Choose negligence a , unobserved by regulator, with benefit $\theta b(a)$
- Opportunity cost relative to a maximum negligence $\theta b(\bar{a})$.

Utility draws a self-reporting cost $c \sim G(\cdot; m)$, worsening in upstream contamination m .

- A test reveals an MCL with probability $q(a)$, $q' \geq 0$.

Model: the utility's decisions to self-report or under-report

Penalties: r for a self-reported exceedance, t for an MR violation, $s > r$ if an inspection (probability p) uncovers a hidden exceedance.

$$\underbrace{c + q(a)r}_{\text{report}} \leq \underbrace{t + pq(a)s}_{\text{under-report}} \implies \text{report iff } c \leq \hat{c}(a) \equiv t - (r - ps)q(a).$$

- A system with **zero** exceedance risk still stops reporting once $c > t$.

$$e(a; c) = \min \{ c + q(a)r, \quad t + q(a)ps \}$$

- The utility chooses a to maximize $-\theta[b(\bar{a}) - b(a)] - e(a; c)$.

Model: why the utility under-reports

Proposition 2

A type- θ utility under-reports if and only if its cost draw exceeds this cutoff, $c > c^*(\theta)$, so the share of utilities under-reporting and receiving an MR violation is:

$$MR(m) = \int_0^{\bar{\theta}} [1 - G(c^*(\theta); m)] f(\theta; m) d\theta \equiv \int_0^{\bar{\theta}} h(\theta, m) f(\theta; m) d\theta.$$

Differentiating splits into two channels:

$$\frac{\partial MR}{\partial m} = \underbrace{\int_0^{\bar{\theta}} \frac{\partial h(\theta, m)}{\partial m} f(\theta; m) d\theta}_{\text{Channel A: cost-distribution shift. Hold } \theta \text{ distribution fixed and change probability type } \theta \text{ utilities draw } c \text{ exceeding } c^*} + \underbrace{\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial f(\theta; m)}{\partial m} d\theta}_{\text{Channel B: type-distribution shift. Hold proportion of type } \theta \text{ utilities whose cutoff exceeds } c^* \text{ fixed and change the proportion of utilities of type } \theta.}$$

Data

- **Violations, enforcement, visits** — SDWIS/ECHO, universe of utilities 1985–2005. Binary by utility $\times$ year; days-in-year as a robustness measure.
 - ▶ Under-inclusive but accurate: 95% of the violations SDWIS holds are correctly reported (US EPA, 2000).
- **Contaminant concentrations** — EPA Six-Year Review round 2, 1998–2005. Cleaned following Ravalli et al. (2022); non-detects set to $MDL/\sqrt{2}$.
- Contaminants: **arsenic, nitrate, barium, selenium** — inorganic chemicals with stable utility MCLs 1985–2005.
- **Mines** — MSHA locations merged to EIA annual production, mine level.
- **Coal sulfur %** — USGS COALQUAL samples, averaged within 20 km of each watershed.
- **Hydrology** — NHDPlus HUC12 watershed boundaries and inflow/outflow.
- **Sample:** Utilities with a coal mine **within two watersheds upstream** of their intake and no coal mine colocated with their intake. 10,641 utility-years, 565 utilities.

Matching intakes to the watershed above them

- HUC12 is the finest watershed mapped for the continental US (>87,000 units).
- Exposure = mines and tons of coal in the HUC12s **within two watersheds upstream** of the utility's intakes.
- Average distance between centroid of adjacent HUC12s is 11 km.
- Intakes **colocated** with a mine are excluded.
- Multiple intakes: sum production, average sulfur.



MR violations are common; almost no MCL violations

Panel A: Any violation in year

Contaminant	MR		MCL	
	%	N	%	N
Nitrates	5.68	612	0.11	12
Arsenic	3.73	402	0.05	5
Inorganic chemicals	4.31	465	0.30	32

Panel B: Days in year

Contaminant	MR				MCL			
	Mean	SD	P90	P99	Mean	SD	P90	P99
Nitrates	14.54	65.35	0.00	365.00	0.27	9.19	0.00	0.00
Arsenic	9.98	54.32	0.00	365.00	0.08	3.83	0.00	0.00
Inorganic chemicals	11.53	58.47	0.00	365.00	0.71	15.08	0.00	0.00

Nitrate MR 5.7%, arsenic 3.7%, IOC 4.3% of utility-years. MCL violations: 12, 5, and 32 events across 1985–2005.

Mining exposure in the sample

Variable	Mean	SD	P90	P99
Number of mines upstream	1.12	2.27	4.00	11.00
Cumul. upstream coal prod. (10M ST) since 1985	0.32	1.00	0.68	5.23
Percentage of coal weight as sulfur	1.11	1.13	2.96	4.01

On average 1.1 mines upstream and 1.1% sulfur by weight; cumulative upstream production reaches 5.2 (10M ST) at the 99th percentile.

Does coal mining contaminate utility water?

Contaminant concentration summary statistics

Variable	MCL	Mean	Max	Std. Dev.	N	Near MCL
Panel A: Full sample						
Arsenic (mg/L)	Pre-2006: 0.050 mg/L, 2006+: 0.010 mg/L	0.0028	0.0110	0.0023	416	0.0%
Nitrate (mg/L)	10.0 mg/L	0.8020	5.5400	0.8738	488	0.4%
Barium (mg/L)	2.000 mg/L	0.0740	0.7071	0.0659	407	0.0%
Selenium (mg/L)	0.050 mg/L	0.0047	0.0354	0.0049	405	0.5%
Panel B: Utility exposed to above median cumulative tons of coal extraction						
Arsenic (mg/L)	Pre-2006: 0.050 mg/L, 2006+: 0.010 mg/L	0.0032	0.0100	0.0026	205	0.0%
Nitrate (mg/L)	10.0 mg/L	0.8254	5.5400	0.8748	260	0.4%
Barium (mg/L)	2.000 mg/L	0.0811	0.3320	0.0569	196	0.0%
Selenium (mg/L)	0.050 mg/L	0.0038	0.0354	0.0043	196	1.0%

Design: cumulative production, measured concentrations

$$Concen_{ct} = \beta \cdot \sum_{\tau=1985}^t \text{UpstreamCoalProd}_{c\tau} + \rho_c + HUC02_c \cdot \tau_t + \epsilon_{ct}$$

- $Concen_{ct}$: mean annual concentration at utility c , year t (SYR2, 1998–2005).
- β : effect of 10 million additional cumulative short tons mined upstream since 1985.
- ρ_c : utility fixed effects. $HUC02_c \cdot \tau_t$: HUC02-specific year effects, absorbing watershed-wide industrial trends.
- SEs clustered at the utility level.
- Continuous treatment, **no control**: identification comes from relative exposure.
- Assumption: utilities with large upstream production changes would have trended like those with small ones. (balance test on utility characteristics $\rightarrow$)

Mining raises contaminant levels

	Arsenic (1)	Nitrate (2)	Barium (3)	Selenium (4)
Panel A: Within one HUC12 upstream				
Cumul. upstream coal prod. (10M ST)	0.0022*** (0.0004)	0.0278 (0.1021)	0.0213** (0.0097)	0.0031** (0.0015)
Panel B: Within two HUC12 upstream				
Cumul. upstream coal prod. (10M ST)	0.0001 (0.0001)	0.0883*** (0.0257)	0.0136 (0.0112)	0.0004** (0.0002)
Observations	366	472	353	350

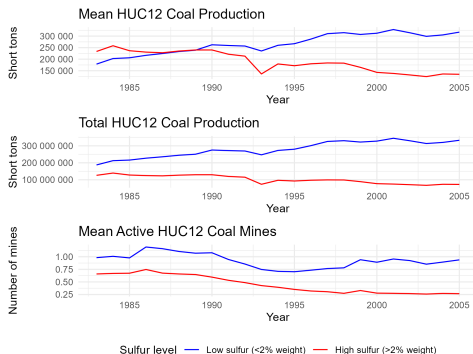
Notes: All specifications include utility and HUC02 $\times$ year fixed effects. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Nitrate +0.09 mg/L per 10M tons (**): 11.0% of its mean. Selenium +0.0004 mg/L (**): 8.5% of its mean. **Every reading in the sample stays under its MCL.** (annual production $\rightarrow$), (number of mines $\rightarrow$)

Do utilities keep self-reporting?

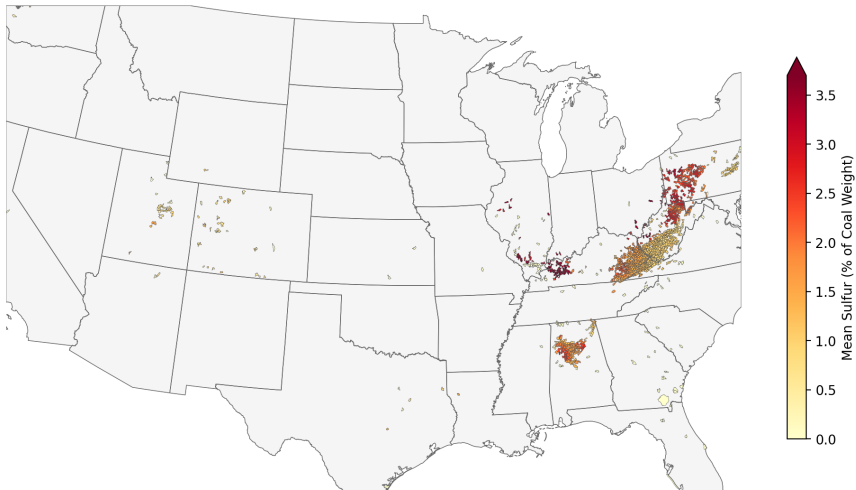
The Acid Rain Program (“ARP”) moved coal production for reasons unrelated to water quality

- Adapt Douglas and Wiggins (2015) ARP instrument for coal production to address endogeneity of mining and utility characteristics.
- 1990 Clean Air Act amendments cap SO_2 from power plants; Phase I deadline **1995**.
- Power plants were 90% of coal demand; switching to low sulfur coal was the main compliance response (Lattanzio, 2022).
- High sulfur mines have limited ability to wash the sulfur out (bound to chemical structure).

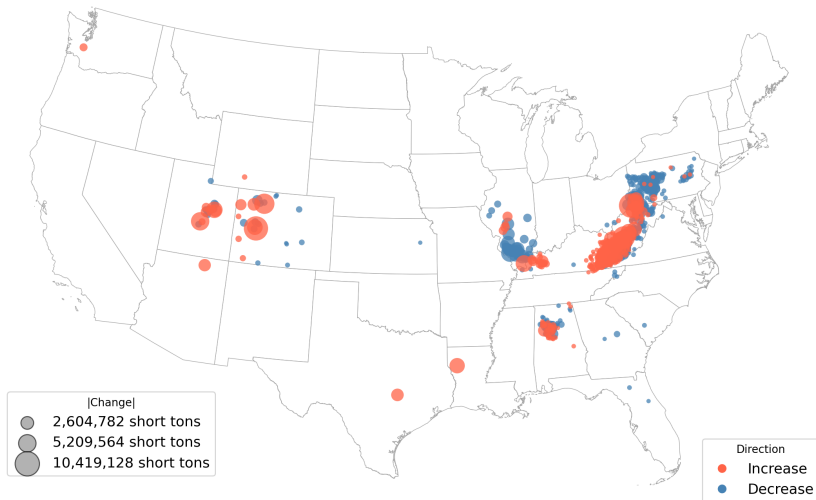


Watershed coal sulfur content

HUC12 watersheds containing coal mines, by mean coal sulfur content



Where coal production moved, 1985–2005



IV design: the ARP as a shifter of where coal is mined

First stage:

$$\text{NumMines}_{ct} = \alpha_0 + \alpha_1 \text{NumFacility}_{ct} + \alpha_2 \text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t \\ + \text{State}_s \cdot Y_t + C_c + \nu_{ct}$$

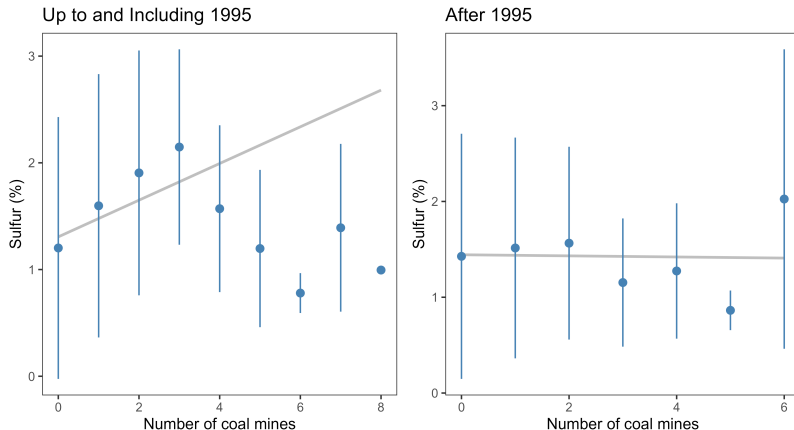
Second stage:

$$\text{Violation}_{vct} = \beta_0 + \beta_1 \text{NumFacility}_{ct} + \gamma \widehat{\text{NumMines}}_{ct} + \text{State}_s \cdot Y_t + C_c + \epsilon_{ct}$$

- $\text{Post95}_t = \mathbf{1}\{t \geq 1995\}$. Sulfur is time-invariant, so C_c absorbs its level.
- Outcome: 1 if utility c had a violation of type v at any point in year t . Coefficients are in percentage points.
- SEs clustered at the utility level.

Shift in location of coal production w.r.t. sulfur

Descriptive evidence of first-stage relevance



Sample: HUC12 watersheds directly upstream of drinking water utility water intakes 1985-2005.

Points show the mean sulfur as a percent of coal weight within each coal-mine-count bin; ranges show ± 1 standard deviation within bin.

First stage: high sulfur watersheds lost their mines

	Upstream coal mines (sum) (1)
Post-1995 $\times$ Upstream sulfur %	-0.4933*** (0.0703)
F-test (1st stage, clustered)	49.20
Utility fixed effects	✓
State $\times$ year fixed effects	✓
Observations	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(first stage for annual upstream coal production $\rightarrow$)

Instrument validity

- **Independence.** The ARP is a national clean-air law set by national energy and emissions politics, not downstream utilities' water quality.
- **Exclusion.** Effect of instrumenting mining downstream of utility on utility behavior is not significant. (upstream and downstream mines instrumented jointly $\rightarrow$); (MR violations $\rightarrow$), (visits $\rightarrow$), (enforcement $\rightarrow$) across k .
Utility characteristics are not predicted by instrument. (full balance table $\rightarrow$)
- **Monotonicity.** A defier is a high-sulfur watershed that expanded *because* its coal was high sulfur. Scrubber adoption costs more likely lead to lost market share rather than gain; low-cost high-sulfur regions that expand anyway are never-takers.

Exclusion test: upstream and downstream mines instrumented jointly

Two endogenous regressors, two instruments:

$$\text{Violation}_{vct} = \gamma_U \widehat{\text{UpMines}}_{ct} + \gamma_D \widehat{\text{DownMines}}_{ct} \\ + \beta_1 \text{NumFacility}_{ct} + \text{State}_s \cdot Y_t + C_c + \epsilon_{ct}$$

- Instruments: $\text{Post95}_t \times \text{UpstreamSulfur}\%_c$ and $\text{Post95}_t \times \text{DownstreamSulfur}\%_c$.
- Downstream mines: coal mines within two watersheds **downstream** of the intake. Their runoff cannot reach the intake.
- If the ARP affects utilities only through upstream mining, $\gamma_D = 0$.
- A nonzero γ_D would signal that the sulfur $\times$ post-1995 shift reaches utilities through another channel (local economy, regulator attention).

Joint first stage: each instrument moves its own mines

	Upstream coal mines (sum) (1)	Downstream coal mines (sum) (2)
Post-1995 $\times$ Upstream sulfur %	-0.4661*** (0.0911)	-0.0845*** (0.0263)
Post-1995 $\times$ Downstream sulfur %	-0.0298 (0.0647)	-0.0534** (0.0227)
F-test (1st stage, clustered)	24.68	18.31
Utility fixed effects	✓	✓
State $\times$ year fixed effects	✓	✓
Observations	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Upstream sulfur predicts upstream mine loss ($F = 24.7$); downstream sulfur predicts downstream mine loss ($F = 18.3$). With utility and year fixed effects: $F = 20.0$ and 17.5.

Downstream mines do not predict MR violations

	Monitoring and reporting (MR) violation					
	Nitrates (1)	(2)	Arsenic (3)	(4)	Inorganic chemicals (5)	(6)
OLS						
Upstream coal mines (sum)	0.54* (0.32)	0.46 (0.34)	0.53* (0.31)	0.37 (0.33)	0.40 (0.31)	0.14 (0.34)
Downstream coal mines (sum)	3.79*** (1.13)	3.82*** (1.17)	2.52** (1.11)	2.35** (1.12)	2.36** (1.11)	2.15* (1.11)
2SLS						
Upstream coal mines (sum)	4.52 (4.74)	2.62 (6.24)	2.69 (3.86)	0.43 (5.08)	1.31 (3.68)	- 1.86 (5.01)
Downstream coal mines (sum)	2.24 (16.28)	4.92 (20.74)	7.97 (13.53)	12.42 (17.16)	12.34 (13.10)	18.19 (17.06)
RF						
Post-1995 × Upstream sulfur %	- 2.20** (1.11)	- 1.63 (1.30)	- 1.78* (0.92)	- 1.25 (1.05)	- 1.47 (0.90)	- 0.67 (1.04)
Post-1995 × Downstream sulfur %	0.08 (0.88)	- 0.34 (0.93)	- 0.24 (0.74)	- 0.68 (0.77)	- 0.49 (0.70)	- 0.92 (0.73)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** p<0.01, ** p<0.05, * p<0.1.

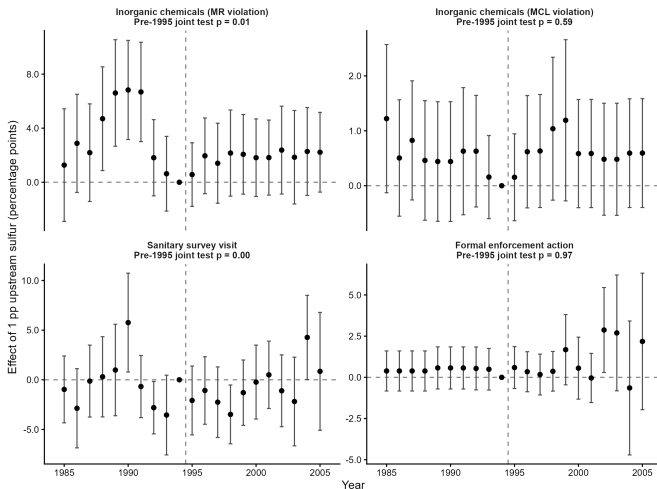
The downstream 2SLS coefficient is insignificant in every column, and also in the combined, MCL, visit, and enforcement tables (29 of 29 columns). Confidence intervals are wide, and the upstream coefficient loses significance too. (combined →), (MCL →), (visits →), (enforcement →), (balance →), (annual production →)

Event study: reduced form by year

$$Y_{ct} = \sum_{s \neq 1994} \delta_s \mathbf{1}\{t = s\} \times \text{UpstreamSulfur}\%_c \\ + \beta_1 \text{NumFacility}_{ct} + \text{State}_s \cdot Y_t + C_c + \epsilon_{ct}$$

- δ_s : effect of 1 pp more upstream sulfur on outcome Y in year s , relative to 1994, the year before ARP Phase I.
- Parallel trends: $\delta_{1985} = \dots = \delta_{1993} = 0$, tested jointly.
- Post – pre: mean of δ_s after 1995 minus mean before, the event-study analogue of the pooled reduced form.

Event study: pre-1995 trends



MCL violations and formal enforcement have flat pre-periods. MR violations rise in 1988–1991 and return to zero by 1994; after 1995 they sit near +2 pp. (MR by contaminant $\rightarrow$), (all outcomes $\rightarrow$)

Event study: joint pre-trend tests, violations

	Pre-1995 joint F	Pre-1995 p-value	Post – pre mean coefficient	Observations
<i>Panel A: Any violation</i>				
Nitrates	2.69	0.00	-1.90** (0.83)	10,641
Arsenic	2.69	0.00	-1.97*** (0.73)	10,641
Inorganic chemicals	2.56	0.01	-1.51* (0.80)	10,641
<i>Panel B: Monitoring and reporting (MR) violation</i>				
Nitrates	3.27	0.00	-1.92** (0.83)	10,641
Arsenic	2.69	0.00	-1.86** (0.73)	10,641
Inorganic chemicals	2.57	0.01	-1.49* (0.79)	10,641
<i>Panel C: Maximum contaminant level (MCL) violation</i>				
Nitrates	0.57	0.82	0.05 (0.09)	10,641
Arsenic	0.15	1.00	-0.11** (0.05)	10,641
Inorganic chemicals	0.83	0.59	0.10 (0.13)	10,641

Notes: All specifications include utility and state $\times$ year fixed effects. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Event study: joint pre-trend tests, visits and enforcement

	Pre-1995 joint F	Pre-1995 p-value	Post – pre mean coefficient	Observations
<i>Panel A: Regulator visit</i>				
Sanitary	4.18	0.00	-0.34 (0.59)	10,641
Technical assistance	1.40	0.19	-0.33 (0.47)	10,641
Enforcement	1.00	0.44	-0.52 (0.39)	10,641
Sample collection	1.25	0.26	-0.22 (0.24)	10,641
Inspection	1.13	0.34	-1.07** (0.49)	10,641
<i>Panel B: Enforcement action</i>				
Informal	2.82	0.00	0.43 (1.04)	10,641
Formal	0.33	0.97	0.55 (0.43)	10,641
None	3.49	0.00	0.12 (1.12)	10,641

Notes: All specifications include utility and state $\times$ year fixed effects. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Mining makes utilities stop reporting

	Monitoring and reporting (MR) violation					
	Nitrates		Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	0.68** (0.33)	0.55 (0.34)	0.62** (0.31)	0.42 (0.33)	0.49 (0.32)	0.19 (0.34)
2SLS						
Upstream coal mines (sum)	5.12*** (1.70)	3.94** (1.75)	4.85*** (1.57)	3.78** (1.55)	4.66*** (1.62)	3.06* (1.66)
RF						
Post-1995 × Upstream sulfur %	-2.12*** (0.66)	-1.95** (0.83)	-2.01*** (0.60)	-1.87** (0.73)	-1.93*** (0.63)	-1.51* (0.79)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Base rates of nitrates, arsenic, and IOC are 5.7%, 3.7%, and 4.3%. OLS estimates are far smaller — consistent with negative omitted variable bias, as expected if poorly performing utilities (more MR violations) are protected from mine siting (fewer mines). (annual upstream coal production →), (event study →)

Reporting cost, not concealment

- Nitrate rules require **more frequent monitoring** once a reading exceeds 50% of the MCL — a discrete jump in c , with no jump in the penalty for under-reporting and no MCL.
- Exceedance risk is held fixed (Channel B = 0), reporting cost increases (Channel A ≥ 0).

	Nitrate MR (1-yr) (1)	Nitrate MR (6-mon) (2)
Concen. > 50% MCL	58.9341** (24.5521)	26.0695** (10.5051)
Mean concen. (z-score)	1.2906 (1.4648)	0.0795 (0.9057)
Observations	835	835

Notes: All specifications include utility and year fixed effects. SEs clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Descriptive, not causal — read the sign, not the magnitude. Utilities under-report when self-reporting gets expensive, not to conceal exceedances.

How do regulators respond?

Enforcement does not counteract under-reporting

	Informal		Enforcement Formal		None	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	-0.15 (0.42)	-0.06 (0.44)	-1.00*** (0.21)	-0.12 (0.19)	0.33 (0.45)	-0.07 (0.48)
2SLS						
Upstream coal mines (sum)	-3.06 (2.15)	-0.71 (2.10)	-3.37*** (0.95)	-1.08 (0.86)	2.37 (2.11)	-0.42 (2.25)
RF						
Post-1995 × Upstream sulfur %	1.26 (0.84)	0.35 (1.03)	1.39*** (0.40)	0.53 (0.41)	-0.98 (0.85)	0.21 (1.11)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

One more upstream mine reduces formal enforcement by 3.4 pp with utility and year fixed effects alone (significant), but the effect loses significance once state × year fixed effects are added. (annual upstream coal production →)

The regulator increases inspections, not enforcement

	Sanitary (1)	Technical assistance (2)	Enforcement (3)	Sample collection (4)	Inspection (5)
OLS					
Upstream coal mines (sum)	0.28 (0.29)	0.10 (0.20)	0.04 (0.11)	-0.07 (0.05)	0.06 (0.13)
2SLS					
Upstream coal mines (sum)	0.78 (1.20)	0.67 (0.95)	1.07 (0.81)	0.48 (0.50)	2.13** (1.07)
RF					
Post-1995 × Upstream sulfur %	-0.39 (0.59)	-0.33 (0.47)	-0.53 (0.39)	-0.24 (0.24)	-1.05** (0.49)
Utility fixed effects	✓	✓	✓	✓	✓
Year fixed effects					
State × year fixed effects	✓	✓	✓	✓	✓
Utilities	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Inspection visits rise significantly (+2.1 pp). Regulators do not systematically step up the visit types tied to sanctions. Visits would reveal concealment and be associated with higher penalties. (annual upstream coal production →)

Conclusion

- Exogenous contamination raises the cost of self-reporting, and self-reporting falls — **even when utilities remain below MCL**.
- Under-reporting due to cost avoidance, rather than concealing a contamination violation.
- Regulators' inspection visits rise significantly — substituting for utility self-reporting.
- Enforcement response is mixed and fixed-effects-sensitive: formal action falls with baseline fixed effects but loses significance once state $\times$ year effects are added.
- **Policy.** Tightening MCLs does nothing if nobody tests. Raising self-reporting penalties could reduce under-reporting.
- Open question: whether regulator monitoring is more efficient than self-reporting.

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Appendix

Appendix: combined MR and MCL violations

	Nitrates		Any violation Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	0.67** (0.33)	0.54 (0.34)	0.60* (0.31)	0.43 (0.34)	0.56* (0.31)	0.28 (0.33)
2SLS						
Upstream coal mines (sum)	5.06*** (1.70)	3.89** (1.75)	4.94*** (1.57)	4.00** (1.56)	4.56*** (1.64)	3.09* (1.67)
RF						
Post-1995 × Upstream sulfur %	-2.09*** (0.66)	-1.92** (0.83)	-2.04*** (0.60)	-1.97*** (0.73)	-1.89*** (0.64)	-1.53* (0.79)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The combined effect mirrors the MR effect: nitrate 3.9–5.1 pp, arsenic 4.0–4.9 pp, IOC 3.1–4.6 pp across the two fixed-effects specifications. (annual upstream coal production →), (MCL violations, annual production →)

Appendix: instrument balance on intake facilities

	Number of Intake Facilities	
	(1)	(2)
Upstream sulfur		0.9942*** (0.3834)
Post-1995 $\times$ Upstream sulfur	0.0509 (0.0386)	0.0563 (0.0376)
Utility fixed effects	✓	
Year fixed effects		✓
State $\times$ year fixed effects	✓	
Balanced panel		✓
Utilities	565	447
Observations	10,641	9,387

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

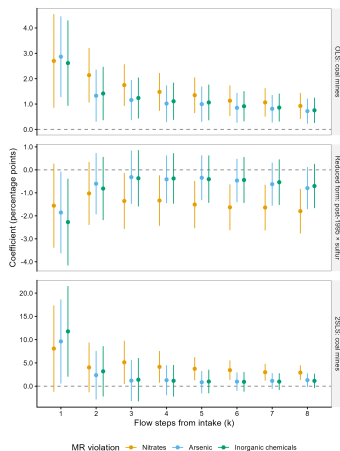
Appendix: balance test on utility characteristics

	Cumul. upstream coal prod. (10M ST)		Any upstream coal mining	
	(1)	(2)	(3)	(4)
Constant	0.2504 (0.2532)		-0.1596 (0.1198)	
Log backcast pop. 1997	0.1091 (0.0750)	0.1512 (0.0938)	0.0159 (0.0126)	0.0208 (0.0144)
Log pop. served (SYR2)	-0.0934 (0.0666)	-0.1113 (0.0956)	0.0060 (0.0259)	-0.0301 (0.0334)
Number of intake facilities	0.0089 (0.0569)	-0.0324 (0.0744)	0.0680* (0.0359)	0.0721* (0.0406)
Surface water source	0.1225 (0.1643)	-0.3902 (0.3296)	0.1210 (0.1307)	0.0521 (0.1601)
Public ownership	-0.1554 (0.2551)	-0.0599 (0.2240)	0.2084** (0.0911)	0.1048 (0.1098)
Wholesaler	-0.3780* (0.2041)	-0.7465 (0.9141)	-0.2164 (0.1340)	-0.2212 (0.1683)
Source water protection	-0.4549 (0.2749)	-0.4385 (0.4305)	-0.2807* (0.1521)	-0.1460 (0.1930)
Joint <i>F</i> -test (all covariates)	2.33	0.96	8.96	2.46
<i>p</i> -value	0.029	0.461	0.000	0.023
Observations	122	120	122	120
State fixed effects		✓		✓

Notes: Heteroskedasticity-robust standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

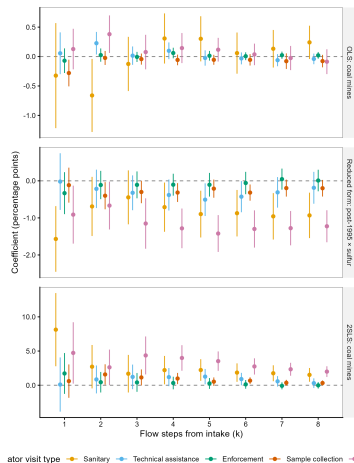
Observable characteristics jointly predict continuous upstream production only without state fixed effects ($p=0.029$) and lose significance once state fixed effects are added ($p=0.461$), as in the main specification. The joint test on the binary presence of upstream mining stays significant with or without state fixed effects.

Appendix: exclusion placebo, MR violations across k



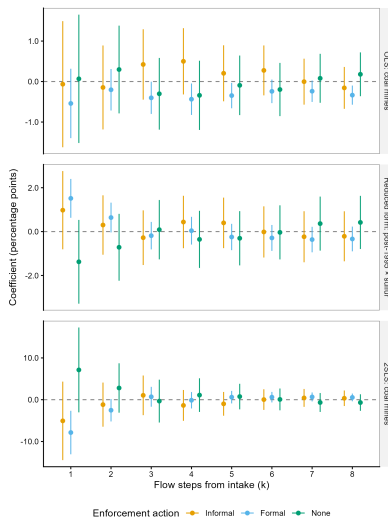
For utilities upstream of a mine — where mining cannot reach the intake hydrologically — OLS estimates on MR violations are positive and significant at every k . In the reduced form and 2SLS, nitrate estimates are significant for $k \geq 3$, while arsenic and inorganic chemical estimates are significant only at $k = 1$.

Appendix: exclusion placebo, visits across k



For utilities upstream of a mine, inspection visits rise significantly at nearly every k , sanitary visits at $k = 1$ and $k \geq 4$, and sample collection visits at several radii; technical assistance and enforcement visits are mostly insignificant.

Appendix: exclusion placebo, enforcement across k



Enforcement actions show no consistent pattern for utilities upstream of a mine at any choice of k , supporting the exclusion restriction.

Robustness: concentrations and annual upstream production

	Arsenic (1)	Nitrate (2)	Barium (3)	Selenium (4)
Panel A: Within one HUC12 upstream				
Upstream coal prod. (1M ST)	0.0032*** (0.0009)	0.0790* (0.0472)	0.0084 (0.0097)	0.0017 (0.0011)
Panel B: Within two HUC12 upstream				
Upstream coal prod. (1M ST)	0.0004 (0.0003)	0.0610*** (0.0153)	0.0137 (0.0123)	0.0004 (0.0004)
Observations	366	472	353	350

Notes: All specifications include utility and HUC02 $\times$ year fixed effects. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Annual production (1M short tons) replaces cumulative production. Arsenic (+0.0032 mg/L, ***) and nitrate (+0.079 mg/L, *) rise within one watershed upstream; nitrate also rises within two (+0.061 mg/L, ***). Barium and selenium are not significant.

Robustness: concentrations and the number of upstream mines

	Arsenic (1)	Nitrate (2)	Barium (3)	Selenium (4)
Panel A: Within one HUC12 upstream				
Upstream coal mines (sum)	0.0000 (0.0001)	0.0472 (0.0495)	0.0035* (0.0018)	0.0001 (0.0002)
Panel B: Within two HUC12 upstream				
Upstream coal mines (sum)	0.0001 (0.0001)	0.0263 (0.0319)	0.0019 (0.0018)	-0.0003 (0.0004)
Observations	366	472	353	350

Notes: All specifications include utility and HUC02 $\times$ year fixed effects. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

With the contemporaneous count of upstream mines as the regressor, only barium within one watershed upstream is marginally significant.

Robustness: first stage for annual upstream coal production

	Upstream coal prod. (1M ST) (1)
Post-1995 $\times$ Upstream sulfur %	-0.1004*** (0.0292)
F-test (1st stage, clustered)	11.79
Utility fixed effects	✓
State $\times$ year fixed effects	✓
Observations	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

High-sulfur watersheds lost 0.10M short tons per percentage point of sulfur after 1995. $F = 11.8$ with state $\times$ year fixed effects, 18.1 with utility and year fixed effects.

Robustness: MR violations and annual upstream production

	Monitoring and reporting (MR) violation					
	Nitrates (1)	(2)	Arsenic (3)	(4)	Inorganic chemicals (5)	(6)
OLS						
Upstream coal prod. (1M ST)	0.97* (0.55)	0.51 (0.54)	1.36** (0.63)	0.54 (0.57)	1.23* (0.65)	0.36 (0.57)
2SLS						
Upstream coal prod. (1M ST)	21.02*** (7.89)	19.39* (9.87)	19.93*** (7.59)	18.58** (9.10)	19.13** (7.66)	15.02* (8.99)
RF						
Post-1995 × Upstream sulfur %	- 2.12*** (0.66)	- 1.95** (0.83)	- 2.01*** (0.60)	- 1.87** (0.73)	- 1.93*** (0.63)	- 1.51* (0.79)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

One million more short tons mined upstream raises MR violations by 15–21 pp, significant in all six columns.

Robustness: combined violations and annual upstream production

	Nitrates		Any violation Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal prod. (1M ST)	0.91* (0.55)	0.48 (0.54)	1.34** (0.63)	0.55 (0.57)	1.25* (0.64)	0.44 (0.57)
2SLS						
Upstream coal prod. (1M ST)	20.75*** (7.86)	19.12* (9.86)	20.27*** (7.64)	19.65** (9.30)	18.74** (7.70)	15.20* (9.05)
RF						
Post-1995 × Upstream sulfur %	- 2.09*** (0.66)	- 1.92** (0.83)	- 2.04*** (0.60)	- 1.97*** (0.73)	- 1.89*** (0.64)	- 1.53* (0.79)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Robustness: MCL violations and annual upstream production

	Maximum contaminant level (MCL) violation					
	Nitrates		Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal prod. (1M ST)	-0.05 (0.06)	-0.01 (0.03)	-0.02 (0.02)	0.01 (0.02)	-0.00 (0.07)	0.03 (0.10)
2SLS						
Upstream coal prod. (1M ST)	-0.35 (0.29)	-0.54 (0.87)	0.34* (0.19)	1.06* (0.58)	-0.65 (1.13)	-1.01 (1.30)
RF						
Post-1995 × Upstream sulfur %	0.04 (0.03)	0.05 (0.09)	-0.03** (0.02)	-0.11** (0.05)	0.07 (0.11)	0.10 (0.13)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

As with mine counts, MCL violations do not respond to upstream production.

Robustness: enforcement and annual upstream production

	Informal		Enforcement Formal		None	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal prod. (1M ST)	- 0.50 (0.94)	0.40 (1.02)	- 1.09*** (0.34)	0.09 (0.33)	0.55 (1.06)	- 0.80 (1.16)
2SLS						
Upstream coal prod. (1M ST)	-12.54 (9.25)	- 3.47 (10.34)	-13.84*** (4.88)	-5.28 (4.38)	9.72 (8.92)	- 2.08 (11.10)
RF						
Post-1995 × Upstream sulfur %	1.26 (0.84)	0.35 (1.03)	1.39*** (0.40)	0.53 (0.41)	-0.98 (0.85)	0.21 (1.11)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Formal enforcement falls by 13.8 pp per million short tons with utility and year fixed effects (***); informal enforcement and no enforcement are not significant.

Robustness: visits and annual upstream production

	Sanitary (1)	Technical assistance (2)	Enforcement (3)	Sample collection (4)	Inspection (5)
OLS					
Upstream coal prod. (1M ST)	0.95 (0.64)	1.25* (0.67)	0.30 (0.46)	0.06 (0.19)	0.51 (0.51)
2SLS					
Upstream coal prod. (1M ST)	3.84 (5.99)	3.27 (4.58)	5.26 (4.21)	2.37 (2.54)	10.46* (5.67)
RF					
Post-1995 $\times$ Upstream sulfur %	-0.39 (0.59)	-0.33 (0.47)	-0.53 (0.39)	-0.24 (0.24)	- 1.05** (0.49)
Utility fixed effects	✓	✓	✓	✓	✓
Year fixed effects					
State $\times$ year fixed effects	✓	✓	✓	✓	✓
Utilities	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Inspection visits rise by 10.5 pp per million short tons (*); other visit types are not significant.

Appendix: joint instruments, combined violations

	Nitrates		Any violation Arsenic		Inorganic chemicals	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	0.54*	0.45	0.51*	0.38	0.47	0.23
	(0.32)	(0.34)	(0.31)	(0.33)	(0.30)	(0.33)
Downstream coal mines (sum)	3.78***	3.82***	2.56**	2.41**	2.27**	2.10*
	(1.13)	(1.17)	(1.11)	(1.12)	(1.10)	(1.10)
2SLS						
Upstream coal mines (sum)	4.80	2.89	2.40	0.27	2.12	- 0.51
	(4.76)	(6.24)	(3.89)	(5.13)	(3.69)	(4.89)
Downstream coal mines (sum)	0.94	3.69	9.34	13.79	9.00	13.33
	(16.38)	(20.78)	(13.68)	(17.37)	(13.10)	(16.58)
RF						
Post-1995 × Upstream sulfur %	- 2.23**	- 1.66	- 1.75*	- 1.29	- 1.60*	- 0.89
	(1.11)	(1.30)	(0.92)	(1.05)	(0.91)	(1.05)
Post-1995 × Downstream sulfur %	0.15	- 0.28	- 0.31	- 0.75	- 0.31	- 0.70
	(0.89)	(0.93)	(0.74)	(0.78)	(0.72)	(0.74)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix: joint instruments, MCL violations

	Maximum contaminant level (MCL) violation					
	Nitrates (1)	(2)	Arsenic (3)	(4)	Inorganic chemicals (5)	(6)
OLS						
Upstream coal mines (sum)	-0.01 (0.01)	-0.00 (0.02)	-0.02 (0.04)	0.01 (0.04)	0.06 (0.07)	0.05 (0.07)
Downstream coal mines (sum)	0.00 (0.02)	0.03 (0.04)	0.04 (0.02)	0.06* (0.03)	-0.10 (0.11)	-0.07 (0.12)
2SLS						
Upstream coal mines (sum)	0.21 (0.45)	0.13 (0.44)	-0.29 (0.26)	-0.15 (0.30)	0.75 (0.78)	0.98 (1.01)
Downstream coal mines (sum)	-1.11 (1.82)	-0.88 (1.96)	1.37 (1.04)	1.37 (1.22)	-3.35 (3.25)	-4.37 (4.22)
RF						
Post-1995 × Upstream sulfur %	-0.02 (0.07)	0.01 (0.07)	0.03 (0.04)	-0.04 (0.05)	-0.10 (0.15)	-0.09 (0.13)
Post-1995 × Downstream sulfur %	0.06 (0.09)	0.04 (0.09)	-0.07 (0.05)	-0.07 (0.05)	0.18 (0.16)	0.20 (0.18)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix: joint instruments, visits

	Sanitary (1)	Technical assistance (2)	Enforcement (3)	Sample collection (4)	Inspection (5)
OLS					
Upstream coal mines (sum)	0.29 (0.30)	0.10 (0.20)	0.04 (0.11)	-0.07 (0.05)	0.04 (0.13)
Downstream coal mines (sum)	- 0.20 (0.84)	- 0.08 (0.29)	-0.07 (0.24)	-0.10 (0.25)	0.48 (0.30)
2SLS					
Upstream coal mines (sum)	- 2.86 (5.20)	0.89 (3.03)	-0.58 (2.74)	-1.46 (2.39)	5.79 (3.86)
Downstream coal mines (sum)	13.49 (18.71)	- 0.83 (11.02)	6.12 (9.04)	7.19 (8.29)	-13.53 (13.65)
RF					
Post-1995 × Upstream sulfur %	0.19 (0.93)	- 0.34 (0.61)	-0.24 (0.57)	0.07 (0.43)	- 1.55** (0.65)
Post-1995 × Downstream sulfur %	- 0.64 (0.80)	0.02 (0.51)	-0.31 (0.38)	-0.34 (0.34)	0.55 (0.51)
Utility fixed effects	✓	✓	✓	✓	✓
Year fixed effects					
State × year fixed effects	✓	✓	✓	✓	✓
Utilities	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(annual production →)

Appendix: joint instruments, enforcement

	Informal		Enforcement Formal		None	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal mines (sum)	- 0.07 (0.42)	- 0.01 (0.44)	-0.95*** (0.21)	-0.10 (0.19)	0.24 (0.45)	- 0.12 (0.48)
Downstream coal mines (sum)	- 2.05* (1.18)	- 1.84 (1.27)	-1.22* (0.71)	-0.66 (0.65)	2.39* (1.24)	2.00 (1.32)
2SLS						
Upstream coal mines (sum)	3.60 (6.26)	7.14 (8.02)	-2.75 (2.63)	1.03 (2.82)	- 1.08 (6.85)	- 5.62 (8.34)
Downstream coal mines (sum)	-24.55 (22.84)	-29.03 (27.98)	-2.28 (9.38)	-7.78 (9.84)	12.73 (24.11)	19.24 (28.79)
RF						
Post-1995 × Upstream sulfur %	0.12 (1.41)	- 0.88 (1.53)	1.41** (0.61)	0.18 (0.57)	- 0.42 (1.58)	1.00 (1.68)
Post-1995 × Downstream sulfur %	1.23 (1.19)	1.34 (1.15)	-0.01 (0.53)	0.39 (0.44)	- 0.61 (1.35)	- 0.86 (1.27)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

(annual production →)

Appendix: joint instruments, balance on intake facilities

	Number of Intake Facilities	
	(1)	(2)
Upstream sulfur		-1.1836 (2.7040)
Downstream sulfur		2.3448 (3.0861)
Post-1995 $\times$ Upstream sulfur	0.0719 (0.0631)	0.0560 (0.0532)
Post-1995 $\times$ Downstream sulfur	-0.0230 (0.0423)	0.0003 (0.0402)
Utility fixed effects	✓	
Year fixed effects		✓
State $\times$ year fixed effects	✓	
Balanced panel		✓
Utilities	565	447
Observations	10,641	9,387

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Robustness: joint instruments, MR violations and annual production

	Monitoring and reporting (MR) violation					
	Nitrates (1)	(2)	Arsenic (3)	(4)	Inorganic chemicals (5)	(6)
OLS						
Upstream coal prod. (1M ST)	0.95* (0.55)	0.51 (0.54)	1.35** (0.63)	0.54 (0.57)	1.22* (0.64)	0.36 (0.57)
Downstream coal prod. (1M ST)	2.62* (1.36)	2.19* (1.32)	1.55 (1.10)	0.97 (1.07)	1.56 (1.11)	1.05 (1.07)
2SLS						
Upstream coal prod. (1M ST)	14.99* (8.45)	13.57 (13.42)	11.94* (6.96)	9.55 (10.70)	9.63 (6.61)	4.05 (10.03)
Downstream coal prod. (1M ST)	14.17 (17.17)	11.40 (18.50)	18.77 (15.37)	17.71 (15.72)	22.31 (15.23)	21.49 (15.08)
RF						
Post-1995 $\times$ Upstream sulfur %	- 2.20** (1.11)	- 1.63 (1.30)	- 1.78* (0.92)	- 1.25 (1.05)	- 1.47 (0.90)	- 0.67 (1.04)
Post-1995 $\times$ Downstream sulfur %	0.08 (0.88)	- 0.34 (0.93)	- 0.24 (0.74)	- 0.68 (0.77)	- 0.49 (0.70)	- 0.92 (0.73)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State $\times$ year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Downstream production is insignificant in all six columns, but the test has low power: the joint first-stage F is 5.5 for downstream production, and 11.3 (utility and year fixed effects) or 5.9 (state $\times$ year fixed effects) for upstream production.

Robustness: joint instruments, enforcement and annual production

	Informal		Enforcement Formal		None	
	(1)	(2)	(3)	(4)	(5)	(6)
OLS						
Upstream coal prod. (1M ST)	- 0.51 (0.94)	0.40 (1.02)	- 1.08*** (0.34)	0.09 (0.33)	0.57 (1.06)	- 0.80 (1.16)
Downstream coal prod. (1M ST)	1.80 (2.77)	1.26 (3.20)	- 1.56 (1.39)	-1.76 (1.40)	- 2.01 (2.60)	- 1.57 (3.08)
2SLS						
Upstream coal prod. (1M ST)	0.09 (10.20)	10.22 (15.19)	- 9.56* (4.92)	-0.81 (5.65)	2.38 (11.52)	-10.33 (16.61)
Downstream coal prod. (1M ST)	-29.66 (22.23)	-26.81 (23.17)	-10.06 (10.92)	-8.77 (8.77)	17.23 (23.02)	16.17 (24.65)
RF						
Post-1995 $\times$ Upstream sulfur %	0.12 (1.41)	- 0.88 (1.53)	1.41** (0.61)	0.18 (0.57)	- 0.42 (1.58)	1.00 (1.68)
Post-1995 $\times$ Downstream sulfur %	1.23 (1.19)	1.34 (1.15)	- 0.01 (0.53)	0.39 (0.44)	- 0.61 (1.35)	- 0.86 (1.27)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State $\times$ year fixed effects		✓		✓		✓
Utilities	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Downstream production is insignificant in all six columns. Joint first-stage F is 5.5 downstream, and 5.9–11.3 upstream.

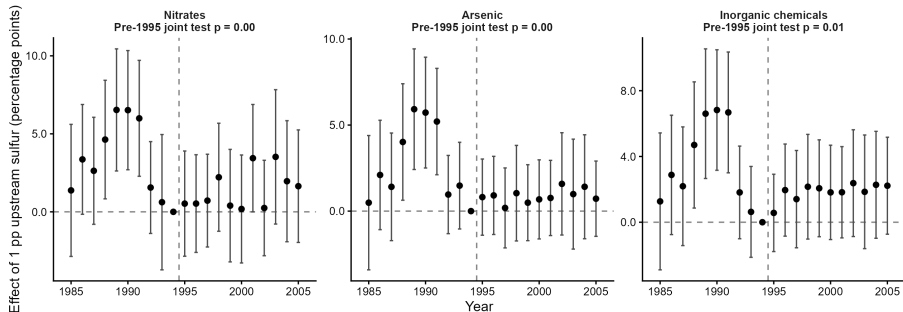
Robustness: joint instruments, visits and annual production

	Sanitary	Technical assistance	Enforcement	Sample collection	Inspection
	(1)	(2)	(3)	(4)	(5)
OLS					
Upstream coal prod. (1M ST)	0.95 (0.64)	1.25* (0.67)	0.30 (0.46)	0.06 (0.19)	0.51 (0.51)
Downstream coal prod. (1M ST)	0.22 (0.91)	1.17 (1.93)	2.71** (1.16)	2.27 (1.70)	4.40*** (1.01)
2SLS					
Upstream coal prod. (1M ST)	- 2.93 (9.58)	3.03 (5.71)	1.53 (5.60)	-1.30 (4.36)	14.59* (8.32)
Downstream coal prod. (1M ST)	13.27 (16.44)	0.47 (9.72)	7.30 (7.53)	7.19 (6.83)	- 8.09 (11.99)
RF					
Post-1995 × Upstream sulfur %	0.19 (0.93)	-0.34 (0.61)	-0.24 (0.57)	0.07 (0.43)	- 1.55** (0.65)
Post-1995 × Downstream sulfur %	- 0.64 (0.80)	0.02 (0.51)	-0.31 (0.38)	-0.34 (0.34)	0.55 (0.51)
Utility fixed effects	✓	✓	✓	✓	✓
Year fixed effects					
State × year fixed effects	✓	✓	✓	✓	✓
Utilities	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

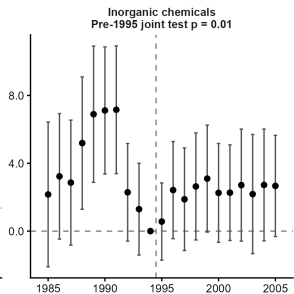
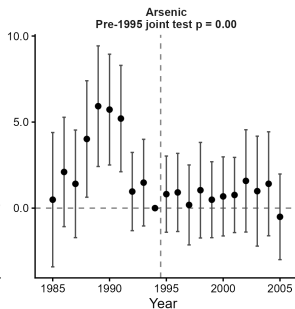
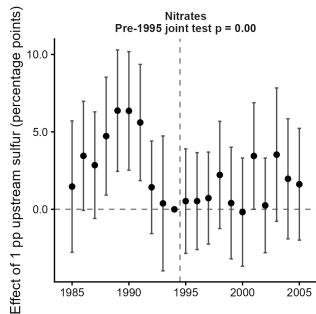
Downstream production is insignificant for every visit type. Joint first-stage F is 5.5 downstream and 5.9 upstream.

Appendix: event study, MR violations by contaminant

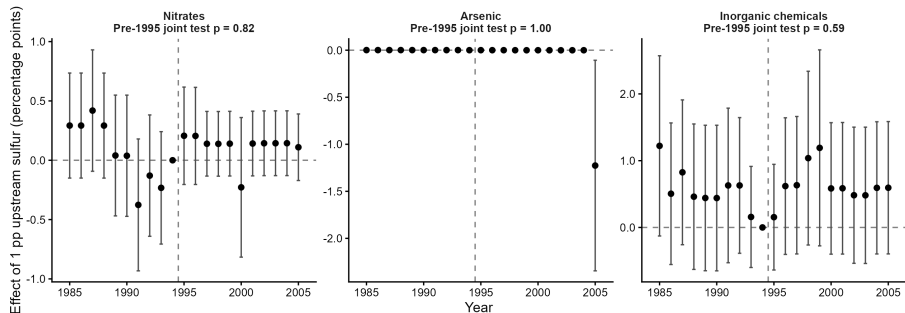


Nitrate, arsenic, and IOC MR violations all rise in 1988–1991 relative to 1994 (pre-1995 joint $p \leq 0.01$) and are flat after 1995.

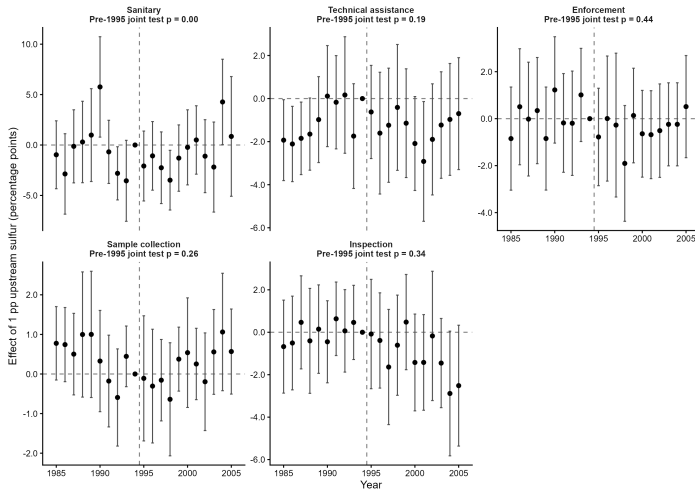
Appendix: event study, any violation



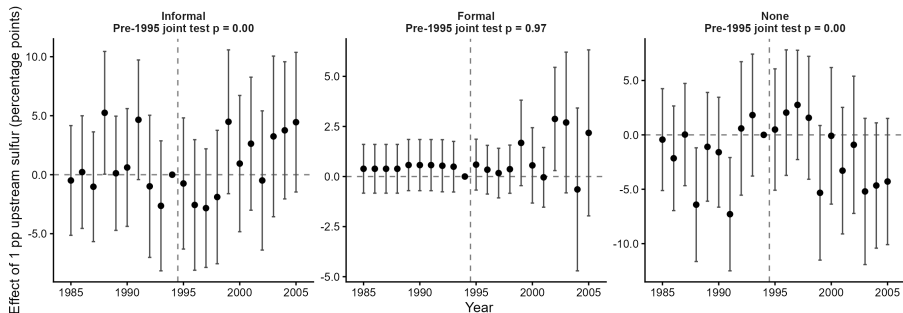
Appendix: event study, MCL violations



Appendix: event study, regulator visits



Appendix: event study, enforcement



Appendix: who appears in the Six-Year Review

Panel A: 1997–2005

	Utility appears in SYR2		Diff.	<i>p</i> -value
	Yes (1)	No (2)		
Ever MR violation	32.2%	18.0%	14.2%***	0.000
Ever MCL violation	1.7%	1.2%	0.5%	0.619
Mean annual violations (count, 0–6)	0.09	0.09	-0.00	0.874

Panel B: 1985–2005

	Utility appears in SYR2		Diff.	<i>p</i> -value
	Yes (1)	No (2)		
Ever MR violation	51.3%	44.1%	7.2%*	0.090
Ever MCL violation	5.5%	3.6%	2.0%	0.277
Mean annual violations (count, 0–6)	0.13	0.20	-0.07**	0.011

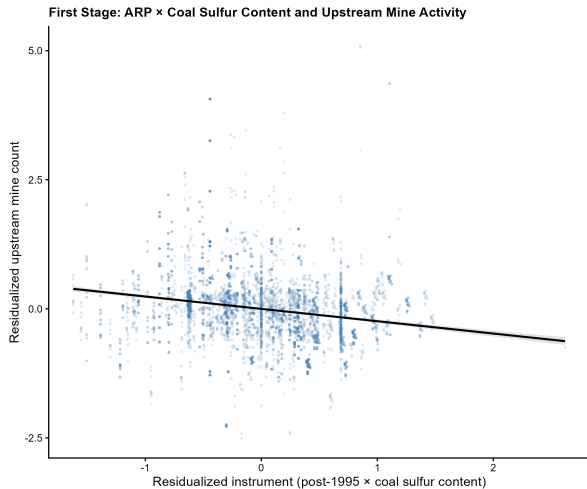
Utilities with SYR2 readings are 7–14 pp more likely to ever incur an MR violation. The concentration sample over-represents the more negligent utilities, so it is not selected toward clean systems.

Appendix: enforcement and visits in the panel

	Whole panel		During MR year		During MCL year	
	%	N	%	N	%	N
Panel A: Enforcement type						
Formal	3.00	323	9.41	70	6.12	3
Informal	18.24	1,965	66.67	496	59.18	29
Any	20.75	2,235	70.70	526	67.35	33
Panel B: Visit type						
Sanitary	13.59	1,464	13.17	98	10.20	5
Technical assistance	2.29	247	4.70	35	2.04	1
Enforcement	1.05	113	2.02	15	4.08	2
Sample collection	0.78	84	0.54	4	0.00	0
Inspection	3.43	369	5.65	42	2.04	1

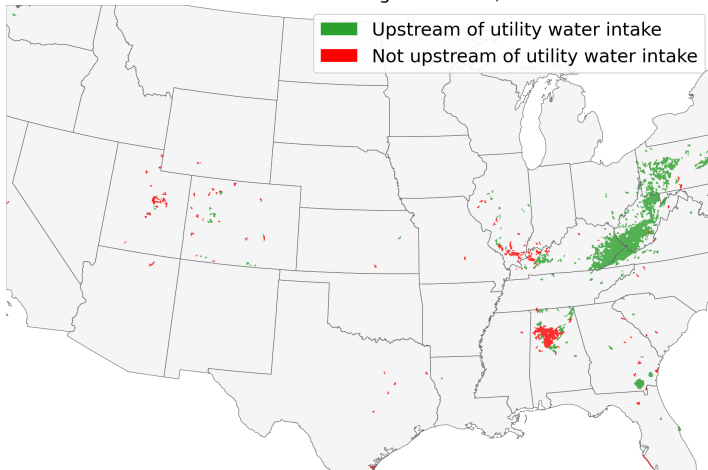
Enforcement responds to violations — any enforcement rises from 21% overall to 71% in MR years — but sanitary visits are no more likely in MR years than in any other year.

Appendix: residualized first stage



Appendix: sample watersheds

HUC12 watersheds containing coal mines, 1985-2005



Appendix: Corollary 1, statement and proof

Write the share of types under-reporting as $h(\theta, m) = 1 - G(c^*(\theta); m)$, with cutoff $c^*(\theta) = t - (r - ps)q(a_{SR}(\theta))$, so that

$$\frac{\partial MR}{\partial m} = \underbrace{\int_0^{\bar{\theta}} \frac{\partial h(\theta, m)}{\partial m} f(\theta; m) d\theta}_{\text{A: cost}} + \underbrace{\int_0^{\bar{\theta}} h(\theta, m) \frac{\partial f(\theta; m)}{\partial m} d\theta}_{\text{B: composition}}.$$

Corollary 1. If $q(a_{SR}(\theta)) \equiv \bar{q}$ on $(0, \bar{\theta})$, then $\partial MR / \partial m = -\partial G(\bar{c}; m) / \partial m \geq 0$, where $\bar{c} = t - (r - ps)\bar{q}$.

Proof. The cutoff is then type-invariant, $c^*(\theta) \equiv \bar{c}$, so $h(\theta, m) = 1 - G(\bar{c}; m)$ does not vary with θ and comes outside both integrals. The limits 0 and $\bar{\theta}$ do not depend on m , so by Leibniz's rule

$$B = [1 - G(\bar{c}; m)] \frac{d}{dm} \left[\int_0^{\bar{\theta}} f(\theta; m) d\theta \right] = [1 - G(\bar{c}; m)] \frac{d}{dm} [1] = 0,$$

since $f(\cdot; m)$ is a density for every m . Channel A collapses to $-\partial G(\bar{c}; m) / \partial m \geq 0$, because contamination shifts reporting costs up. $\square$

Why it matters. Channel B is the concealment margin: it operates only if contamination reallocates mass toward types whose exceedance risk, and hence whose cutoff, differs. Shut it down and any remaining rise in under-reporting is avoidance of the reporting technology itself. The 50%-of-MCL nitrate step is the empirical counterpart — monitoring frequency jumps, the sanction schedule does not, and readings are nowhere near the limit.

Robustness: mine accident instruments

Mine accident instruments: estimating equations

Second stage:

$$\text{Violation}_{vct} = \gamma_U \widehat{\text{UpProd}}_{ct} + \gamma_D \widehat{\text{DownProd}}_{ct} \\ + \beta_1 \text{NumFacility}_{ct} + \text{State}_s \cdot Y_t + C_c + \epsilon_{ct}$$

First stage, for each $P \in \{\text{UpProd}, \text{DownProd}\}$:

$$P_{ct} = \pi_1^P \text{UpAcc}_{ct} + \pi_2^P \text{UpSulfur}\%_c \cdot \text{Post95}_t \\ + \pi_3^P \text{DownAcc}_{ct} + \pi_4^P \text{DownSulfur}\%_c \cdot \text{Post95}_t \\ + \alpha_1 \text{NumFacility}_{ct} + \text{State}_s \cdot Y_t + C_c + \nu_{ct}$$

- Prod: annual coal production (1M short tons) within two watersheds upstream / downstream of the intake. Acc: MSHA accident, injury, and illness reports at those mines (units of 10).
- Columns add instruments in order: (1) UpAcc; (2) + UpSulfur $\times$ Post95; (3) + DownAcc, with DownProd as a second endogenous regressor; (4) + DownSulfur $\times$ Post95.
- Outcomes are 0/1; coefficients in percentage points. SEs clustered at the utility level.

Mine accident instruments: MR violations

	Nitrates				Monitoring and reporting (MR) violation Arsenic				Inorganic chemicals			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: 2SLS												
Upstream coal prod. (1M ST)	- 0.38 (1.47)	0.79 (1.52)	0.65 (1.54)	0.63 (1.54)	- 0.95 (1.31)	0.21 (1.36)	- 0.05 (1.37)	- 0.09 (1.37)	- 1.11 (1.50)	- 0.15 (1.56)	- 0.42 (1.57)	- 0.47 (1.57)
Downstream coal prod. (1M ST)			4.96 (4.82)	5.08 (4.79)			6.97 (4.89)	7.22 (4.88)			6.58 (4.92)	6.99 (4.93)
Panel B: First stage												
Upstream coal prod. (1M ST)												
Upstream accident reports (10s)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)
Post-1995 × Upstream sulfur %		- 0.07*** (0.03)	- 0.08*** (0.03)	- 0.09*** (0.03)		- 0.07*** (0.03)	- 0.08*** (0.03)	- 0.09*** (0.03)		- 0.07*** (0.03)	- 0.08*** (0.03)	- 0.09*** (0.03)
Downstream accident reports (10s)			- 0.02** (0.01)	- 0.02** (0.01)			- 0.02** (0.01)	- 0.02** (0.01)			- 0.02** (0.01)	- 0.02** (0.01)
Post-1995 × Downstream sulfur %				0.02 (0.02)				0.02 (0.02)				0.02 (0.02)
First-stage F	77.03	57.00	39.95	31.63	77.03	57.00	39.95	31.63	77.03	57.00	39.95	31.63
Downstream coal prod. (1M ST)												
Upstream accident reports (10s)			0.00 (0.00)	0.00 (0.00)			0.00 (0.00)	0.00 (0.00)			0.00 (0.00)	0.00 (0.00)
Post-1995 × Upstream sulfur %			- 0.04*** (0.01)	- 0.01 (0.01)			- 0.04*** (0.01)	- 0.01 (0.01)			- 0.04*** (0.01)	- 0.01 (0.01)
Downstream accident reports (10s)			0.12*** (0.02)	0.12*** (0.02)			0.12*** (0.02)	0.12*** (0.02)			0.12*** (0.02)	0.12*** (0.02)
Post-1995 × Downstream sulfur %				- 0.03*** (0.01)				- 0.03*** (0.01)				- 0.03*** (0.01)
First-stage F			11.20	8.51			11.20	8.51			11.20	8.51
Utilities	565	565	565	565	565	565	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Upstream accidents strongly predict upstream production ($F = 77.0$ in column 1). With accidents in the instrument set, the upstream 2SLS estimate on MR violations is near zero and insignificant (IOC -1.11 to -0.47), well below the sulfur-only estimate of 15.02. Downstream production is insignificant; its first-stage F falls to 8.5 in column 4.

Mine accident instruments: enforcement

	Informal				Enforcement Formal				None			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: 2SLS												
Upstream coal prod. (1M ST)	2.27 (2.57)	1.93 (2.57)	1.95 (2.57)	2.07 (2.57)	1.71* (1.01)	1.30 (0.94)	1.56 (1.00)	1.57 (1.00)	- 2.65 (2.77)	- 2.62 (2.81)	- 2.53 (2.80)	- 2.61 (2.80)
Downstream coal prod. (1M ST)			- 1.13 (6.00)	- 1.92 (6.06)			- 5.37 (3.59)	- 5.45 (3.55)			- 1.41 (6.29)	- 0.85 (6.29)
Panel B: First stage												
Upstream coal prod. (1M ST)												
Upstream accident reports (10s)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)	0.12*** (0.01)
Post-1995 × Upstream sulfur %		- 0.07*** (0.03)	- 0.08*** (0.03)	- 0.09*** (0.03)		- 0.07*** (0.03)	- 0.08*** (0.03)	- 0.09*** (0.03)		- 0.07*** (0.03)	- 0.08*** (0.03)	- 0.09*** (0.03)
Downstream accident reports (10s)			- 0.02** (0.01)	- 0.02** (0.01)			- 0.02** (0.01)	- 0.02** (0.01)			- 0.02** (0.01)	- 0.02** (0.01)
Post-1995 × Downstream sulfur %				0.02 (0.02)				0.02 (0.02)				0.02 (0.02)
First-stage F	77.03	57.00	39.95	31.63	77.03	57.00	39.95	31.63	77.03	57.00	39.95	31.63
Downstream coal prod. (1M ST)												
Upstream accident reports (10s)			0.00 (0.00)	0.00 (0.00)			0.00 (0.00)	0.00 (0.00)			0.00 (0.00)	0.00 (0.00)
Post-1995 × Upstream sulfur %			- 0.04*** (0.01)	- 0.01 (0.01)			- 0.04*** (0.01)	- 0.01 (0.01)			- 0.04*** (0.01)	- 0.01 (0.01)
Downstream accident reports (10s)			0.12*** (0.02)	0.12*** (0.02)			0.12*** (0.02)	0.12*** (0.02)			0.12*** (0.02)	0.12*** (0.02)
Post-1995 × Downstream sulfur %				- 0.03*** (0.01)				- 0.03*** (0.01)				- 0.03*** (0.01)
First-stage F			11.20	8.51			11.20	8.51			11.20	8.51
Utilities	565	565	565	565	565	565	565	565	565	565	565	565
Observations	10,641	10,641	10,641	10,641	10,641	10,641	10,641	10,641	10,641	10,641	10,641	10,641

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Upstream production raises formal enforcement only in column 1 (1.71 pp, *). No other upstream or downstream coefficient is significant.

Mine accident instruments: regulator visits

[illegible]

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Upstream production has no significant effect on any visit type. Downstream production raises inspection visits (5.01 and 4.56 pp, **) and enforcement visits in column 4 (1.66 pp, *).

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